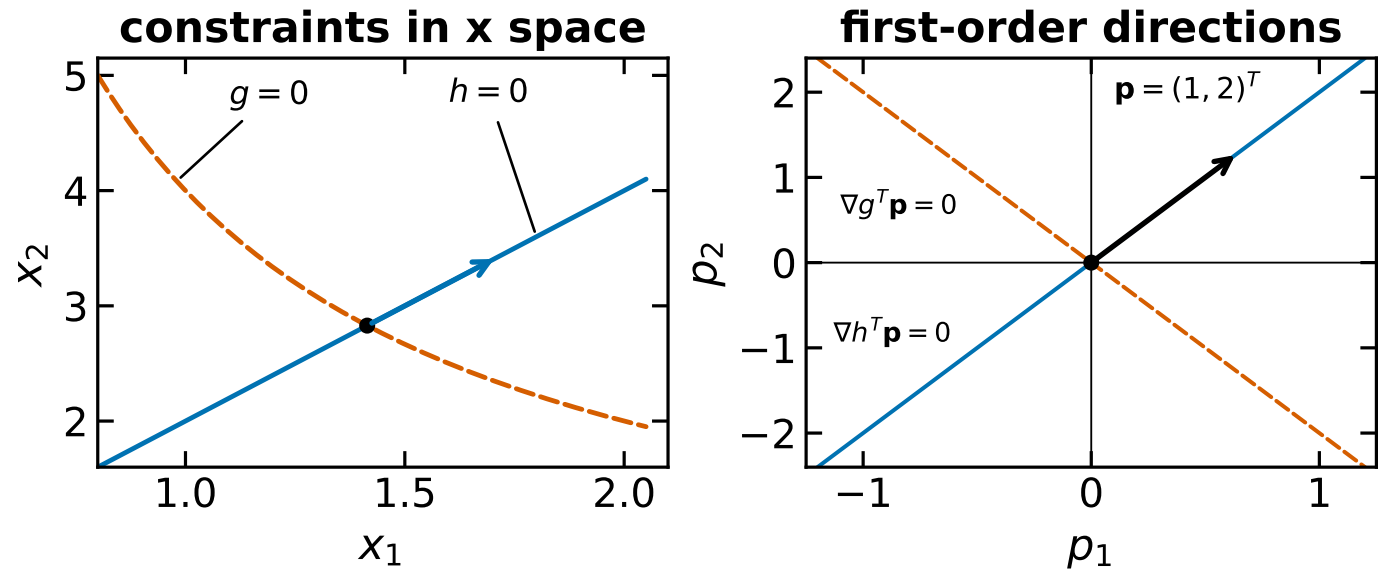
constraints in x space
$g = 0$
$h = 0$
x_2
x_1
5
4
3
2
1.0
1.5
2.0
first-order directions
$\mathbf{p} = (1, 2)^T$
$\nabla g^T \mathbf{p} = 0$
$\nabla h^T \mathbf{p} = 0$
p_2
p_1
2
1
0
−1
−2
−1
0
1